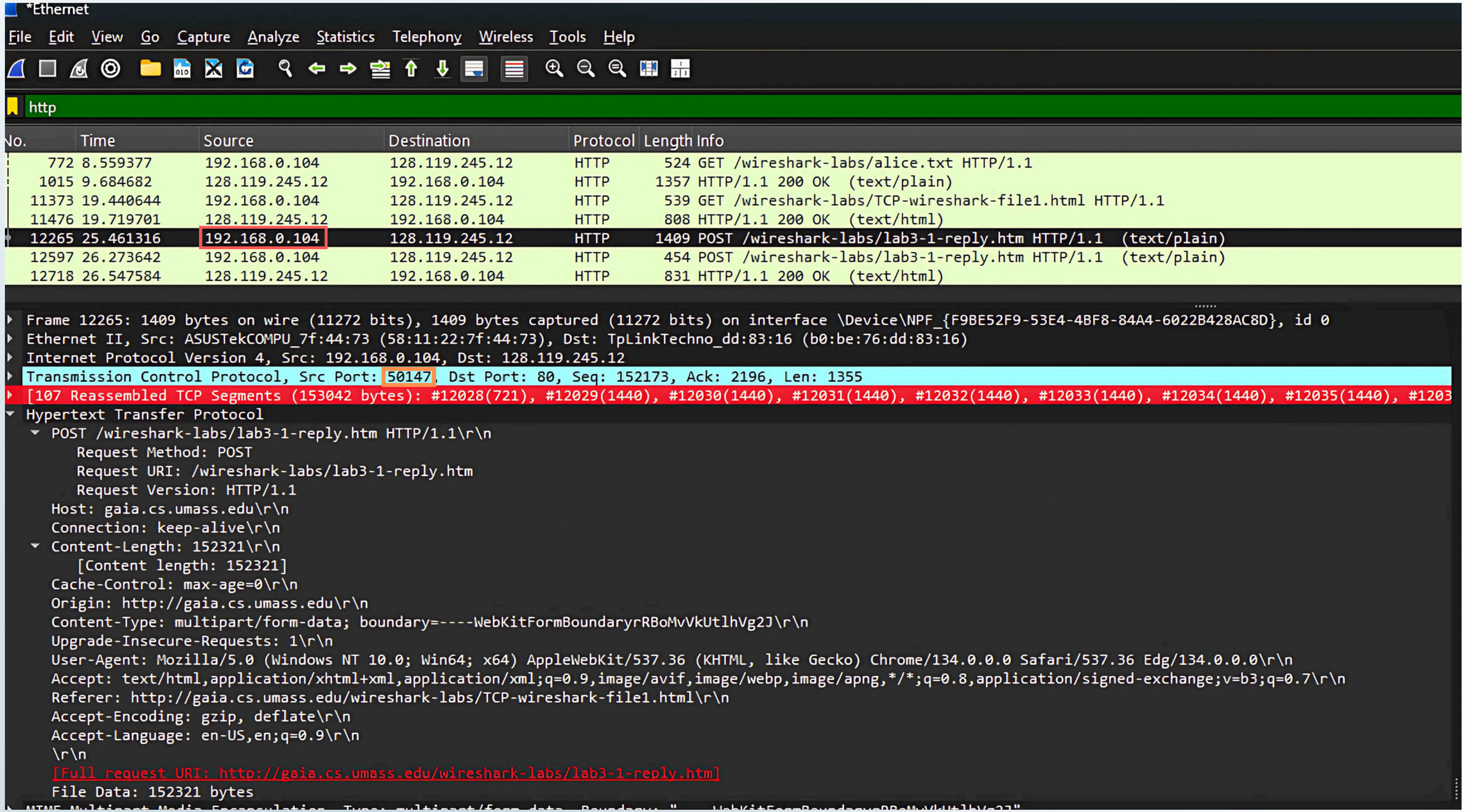
จุติณัฏฐ์ รัตนะมงคลกุล 6609650228
รพินทร์ นะราช 6609650624

Wireshark
Lab 04 -- TCP
Assignment

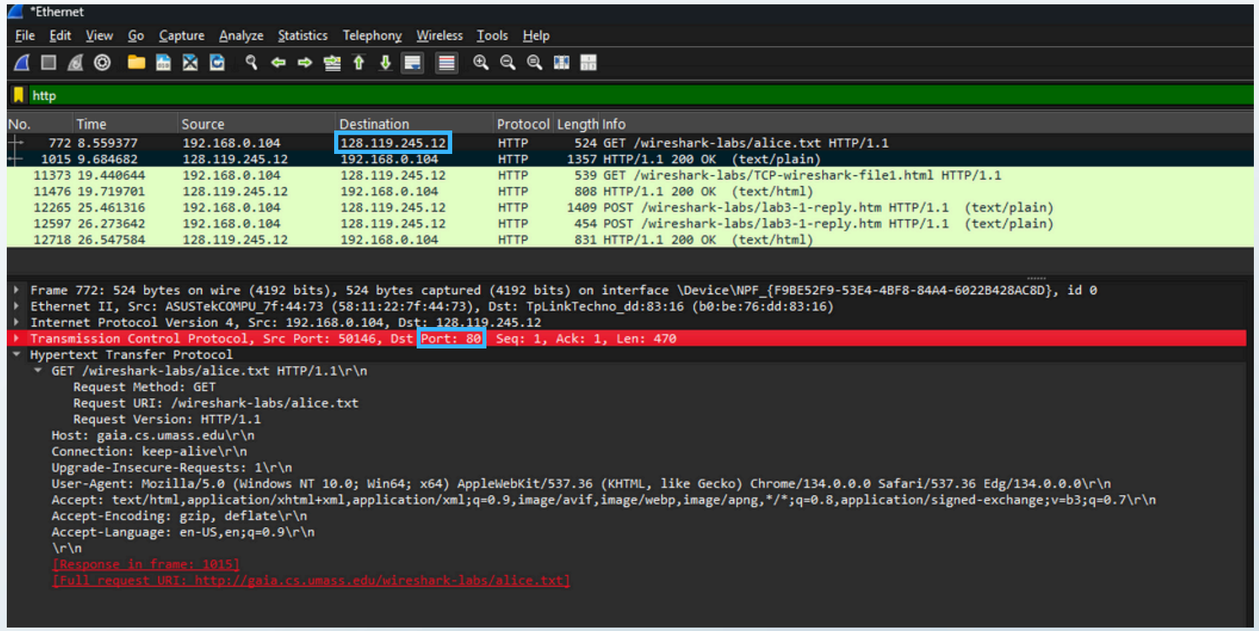


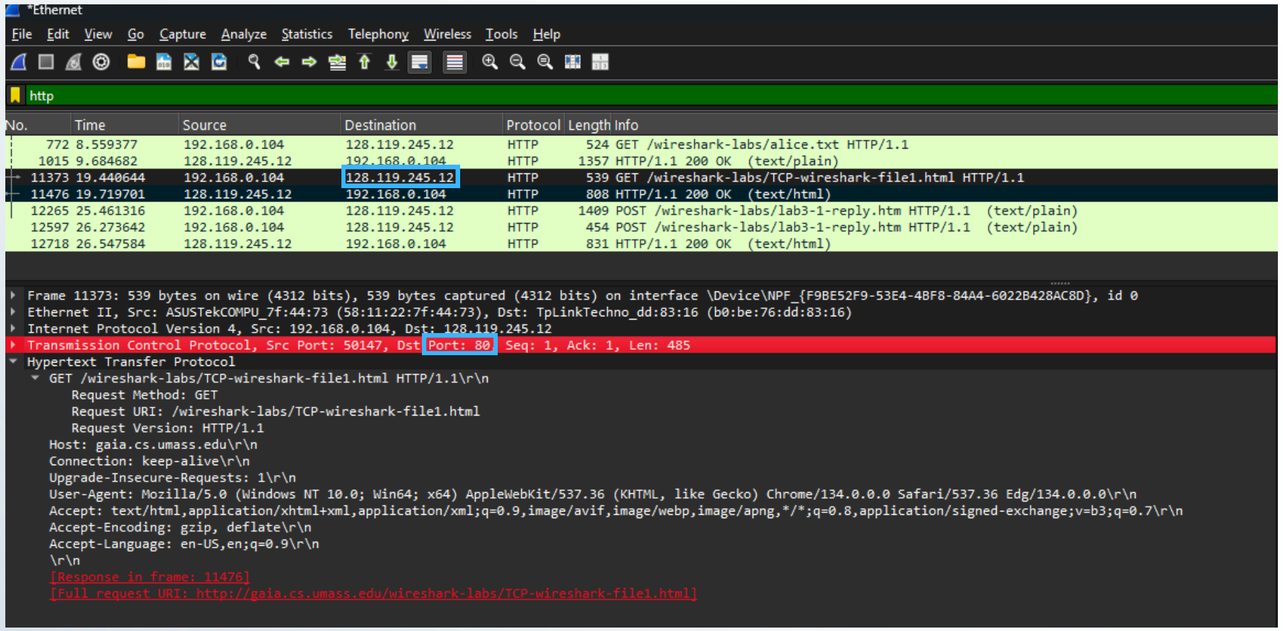
1. What is the IP address and TCP port number used by the client computer (source) that is transferring the alice.txt file to gaia.cs.umass.edu? To answer this question, it's probably easiest to select an HTTP message and explore the details of the TCP packet used to carry this HTTP message, using the "details of the selected packet header window" (refer to Figure 2 in the "Getting Started with Wireshark" Lab if you're uncertain about the Wireshark windows).

Source IP คือ 192.168.0.104
Source Port คือ 50147



2. What is the IP address of gaia.cs.umass.edu? On what port number is it sending and receiving TCP segments for this connection?





IP address ของ gaia.cs.umass.edu : 128.119.245.12

Destination Port : 80

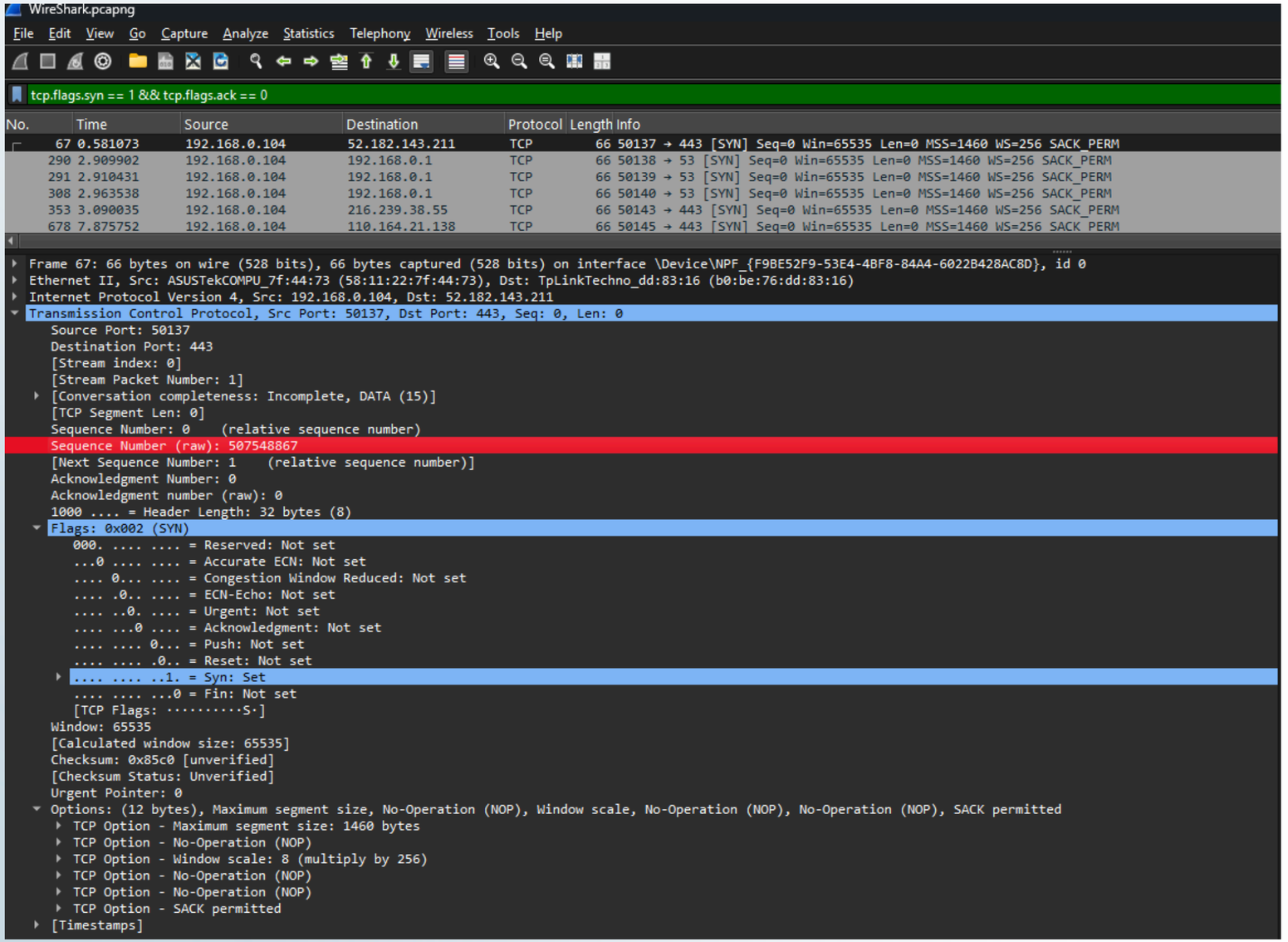
3. What is the sequence number of the TCP SYN segment that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu? (Note: this is the “raw” sequence number carried in the TCP segment itself; it is NOT the packet # in the “No.” column in the Wireshark window. Remember there is no such thing as a “packet number” in TCP or UDP; as you know, there are sequence numbers in TCP and that’s what we’re after here. Also note that this is not the relative sequence number with respect to the starting sequence number of this TCP session.). What is it in this TCP segment that identifies the segment as a SYN segment? Will the TCP receiver in this session be able to use Selective Acknowledgments (allowing TCP to function a bit more like a “selective repeat” receiver, see section 3.4.5 in the text)?

Sequence Number : 507548867

Relative Sequence Number : 1

แพ็กเก็ตนี้เป็น SYN เพราะมี SYN Flag ถูกตั้งค่า

TCP Receiver รองรับ SACK เพราะ ใน TCP Option มี Sack Permitted



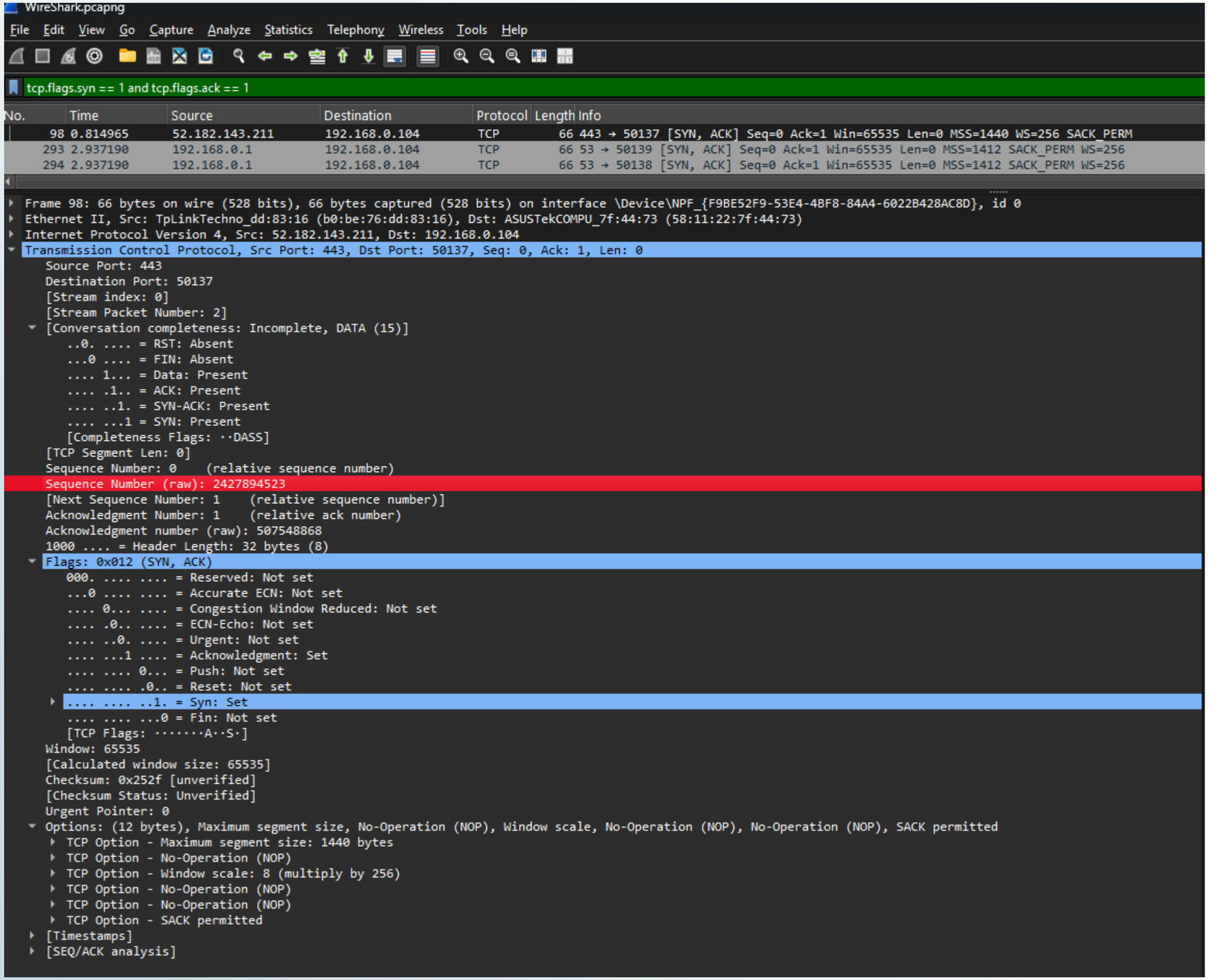
4. What is the sequence number of the SYNACK segment sent by gaia.cs.umass.edu to the client computer in reply to the SYN? What is it in the segment that identifies the segment as a SYNACK segment? What is the value of the Acknowledgement field in the SYNACK segment? How did gaia.cs.umass.edu determine that value?

-แพ็กเก็ต No.98

-Sequence Number : 2427894523

-Sequence Number : 0 (relative sequence number)

-Acknowledgment Number : 507548868



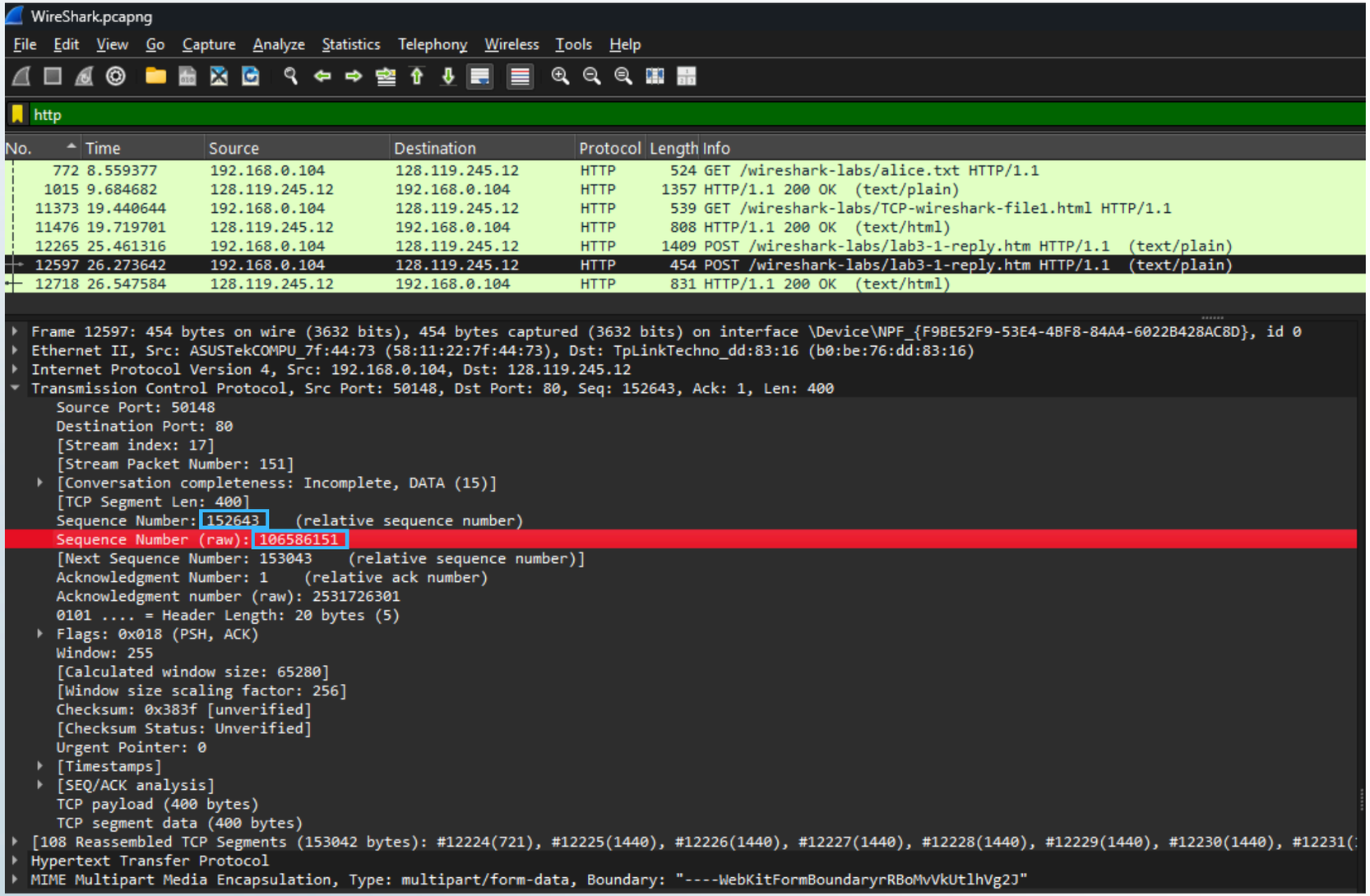
5. What is the sequence number of the TCP segment containing the header of the HTTP POST command? Note that in order to find the POST message header, you'll need to dig into the packet content field at the bottom of the Wireshark window, looking for a segment with the ASCII text "POST" within its DATA field4,5 . How many bytes of data are contained in the payload (data) field of this TCP segment? Did all of the data in the transferred file alice.txt fit into this single segment?

Sequence Number (raw): 106586151'

Sequence Number: 152643 (relative sequence number)

TCP payload: 400 bytes

ข้อมูล alice.txt ไม่ได้อยู่ใน segment เดียว



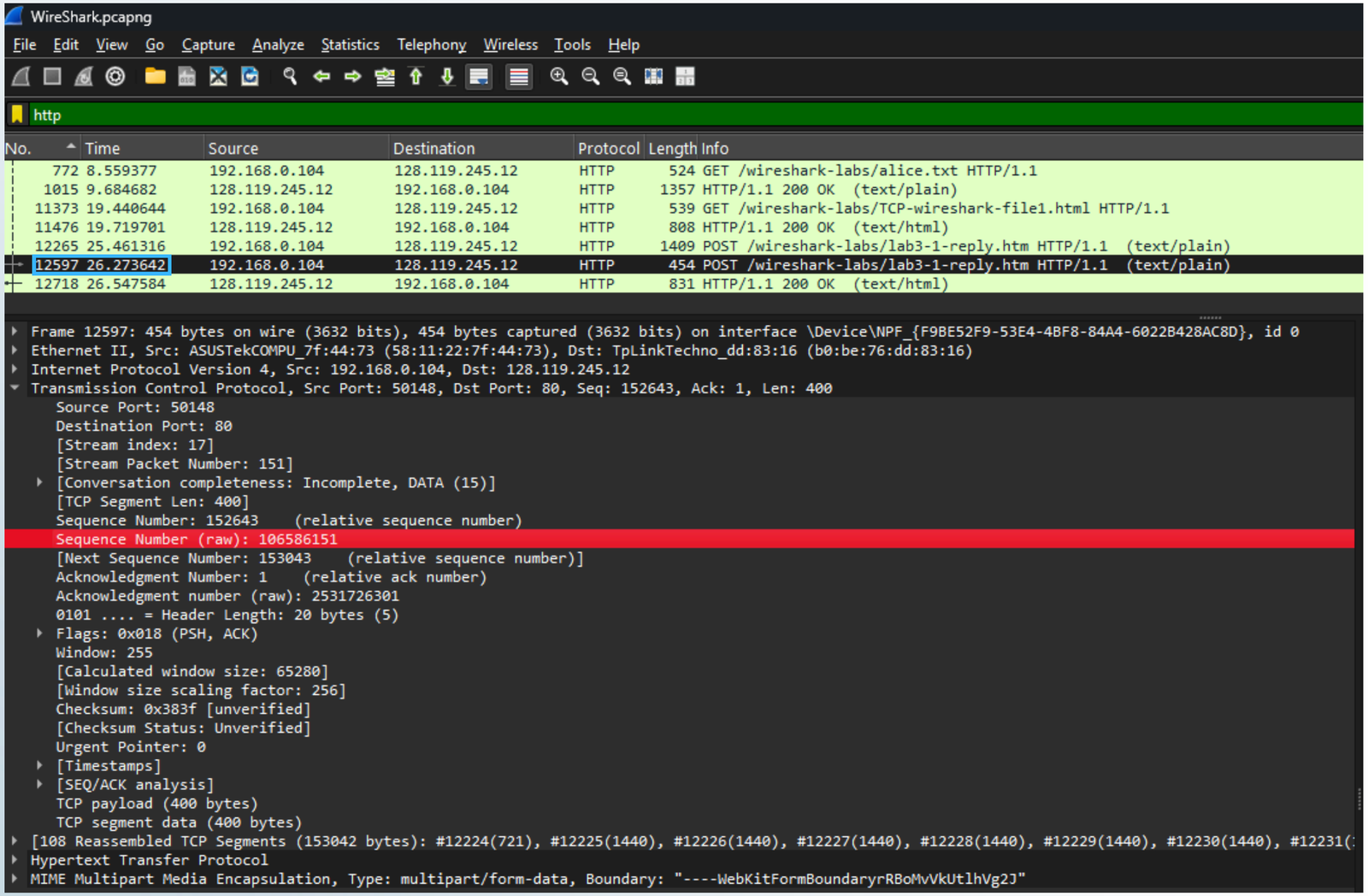
6. Consider the TCP segment containing the HTTP “POST” as the first segment in the data transfer part of the TCP connection.

- At what time was the first segment (the one containing the HTTP POST) in the data-transfer part of the TCP connection sent?
- At what time was the ACK for this first data-containing segment received?
- What is the RTT for this first data-containing segment?
- What is the RTT value the second data-carrying TCP segment and its ACK?
- What is the EstimatedRTT value (see Section 3.5.3, in the text) after the ACK for the second data-carrying segment is received? Assume that in making this calculation after the received of the ACK for the second segment, that the initial value of EstimatedRTT is equal to the measured RTT for the first segment, and then is computed using the EstimatedRTT equation on page 242, and a value of $\alpha = 0.125$.

Note: Wireshark has a nice feature that allows you to plot the RTT for each of the TCP segments sent. Select a TCP segment in the “listing of captured packets” window that is being sent from the client to the gaia.cs.umass.edu server. Then select: Statistics->TCP Stream Graph- >Round Trip Time Graph.

6.1 At what time was the first segment (the one containing the HTTP POST) in the data-transfer part of the TCP connection sent?

Timestamp ของ แพ็กเก็ต No.12597 และ เวลา 26.273642 วินาที

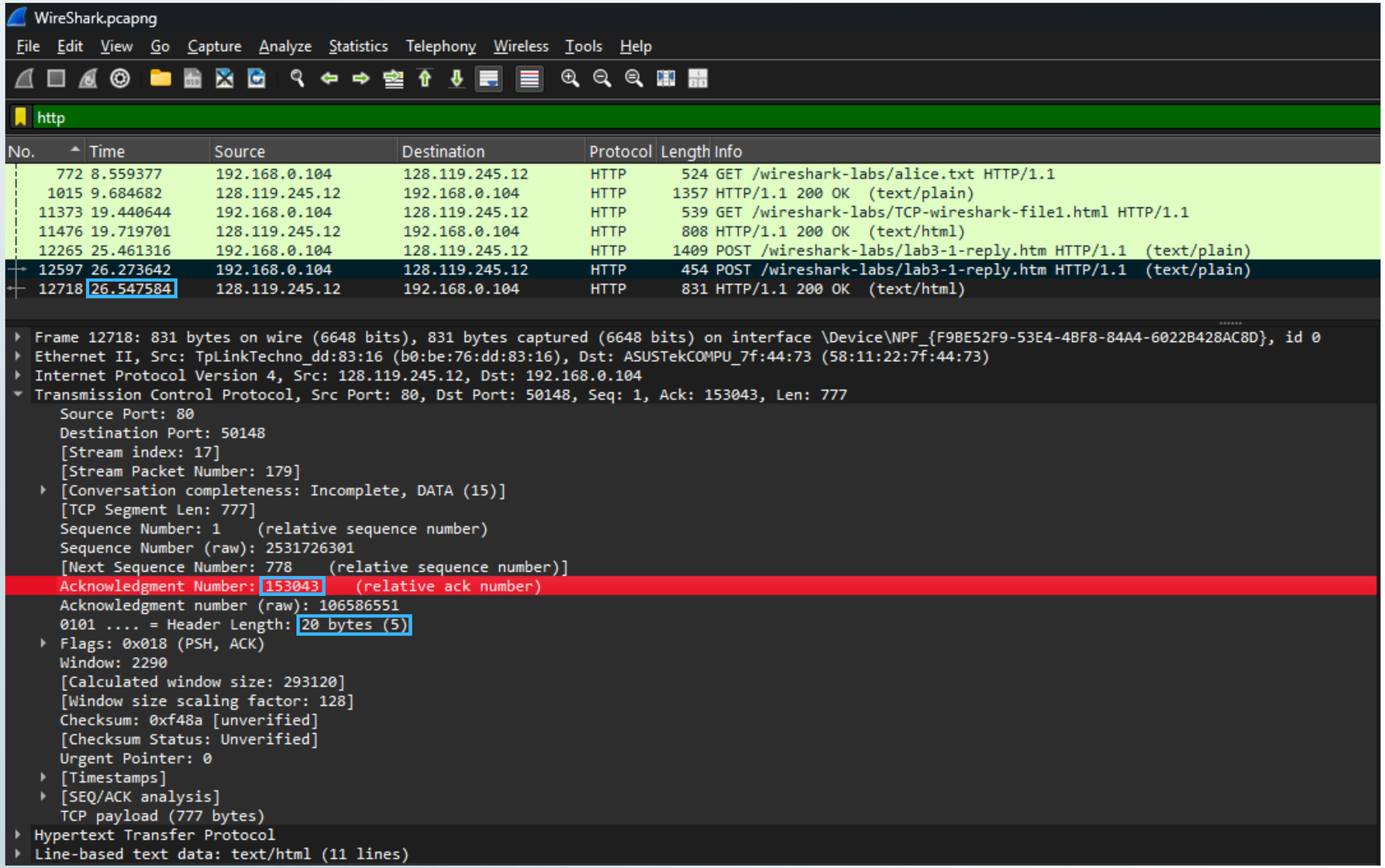


6.2 At what time was the ACK for this first data-containing segment received?

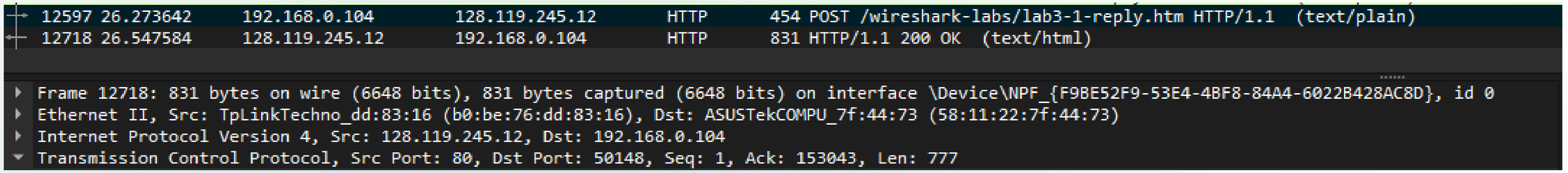
เวลาหลังจากที่ส่ง POST ไป เซิร์ฟเวอร์จะตอบกลับด้วย

ACK = Seq + Len ของแพ็กเก็ต POST :
Seq + Len ของแพ็กเก็ต
[No.12597] = 152643 + 400 = 153043

ACK Number ของแพ็กเก็ต
[No.12718] = 153043
เวลาที่ ACK ถูกส่งกลับ = 26.547584



6.3 What is the RTT for this first data-containing segment?



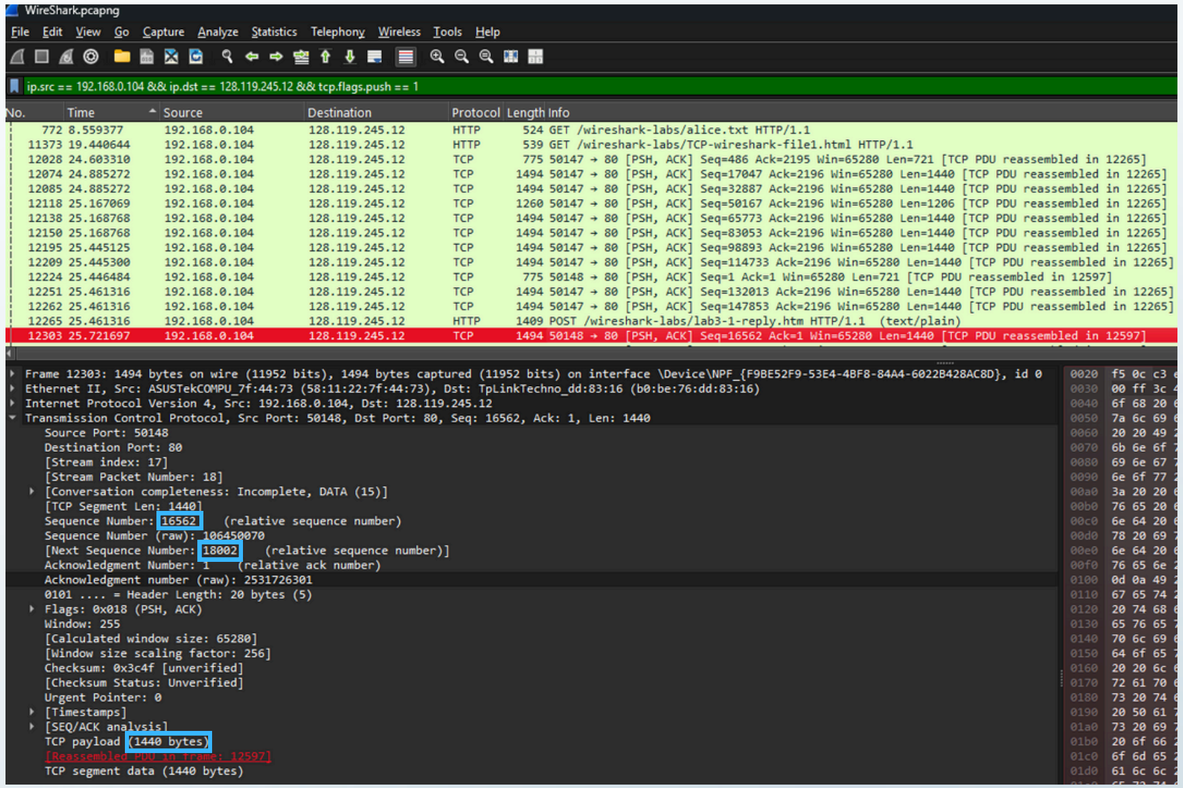
RTT สำหรับแพ็กเก็ตแรก = (เวลาของ ACK) - (เวลาของแพ็กเก็ตแรก)

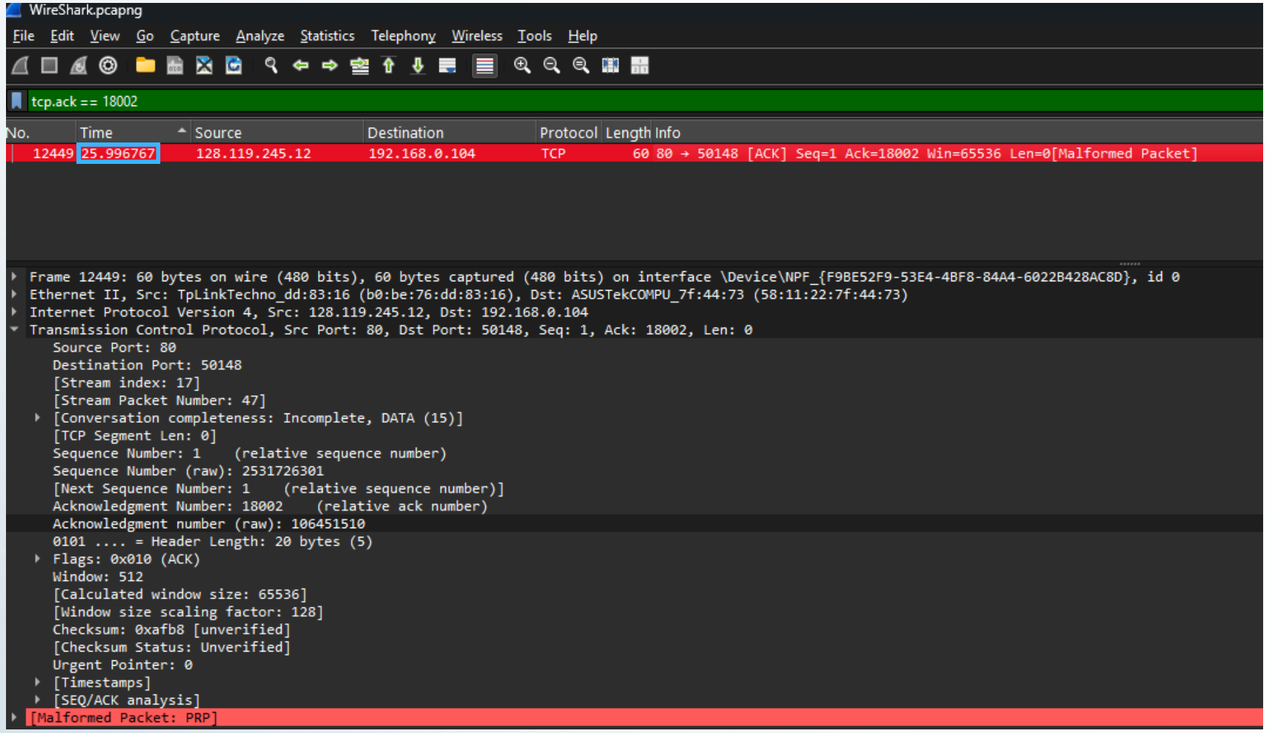
26.547584 (No. 12597)

\- 26.273642 (No. 12718)

0.273942 วินาที

6.4 What is the RTT value the second data-carrying TCP segment and its ACK?





ACK ที่ควรได้รับ = Sequence Number + Payload Length

16562 + 1440 = 18002

RTT = (เวลาของ ACK ที่ส่งกลับ) - (เวลาของแพ็กเก็ตสอง)

26.547584 - 25.996767 = 0.550817 วินาที

6.5 What is the EstimatedRTT value (see Section 3.5.3, in the text) after the ACK for the second data-carrying segment is received? Assume that in making this calculation after the received of the ACK for the second segment, that the initial value of EstimatedRTT is equal to the measured RTT for the first segment, and then is computed using the EstimatedRTT equation on page 242, and a value of $\alpha = 0.125$.

ใช้ $EstimatedRTT = (1 - \alpha) \times EstimatedRTT_{previous} + \alpha \times RTT_{sample}$

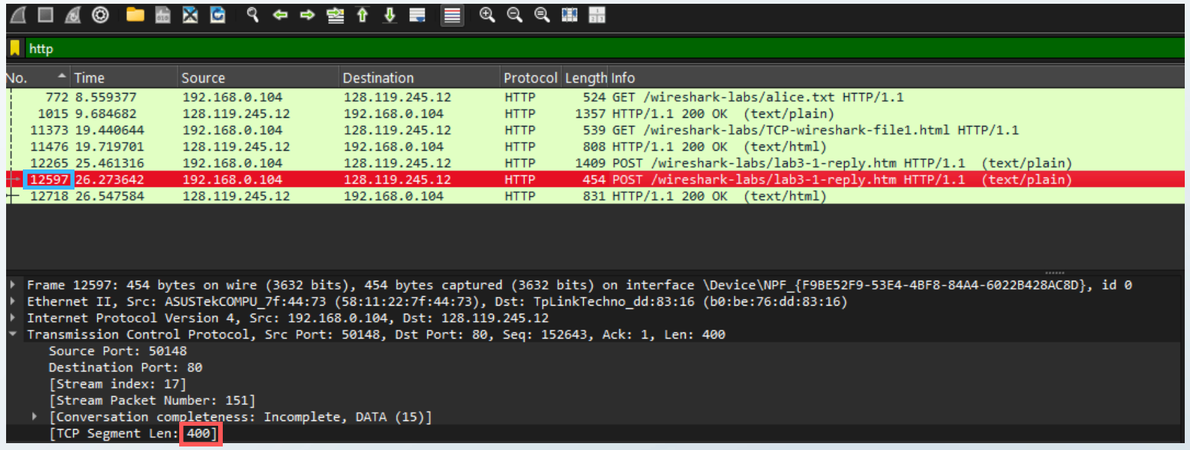
EstimatedRTT = (1 - 0.125) × RTT(เซกเมนต์แรก) + 0.125 × RTT(เซกเมนต์ที่สอง)

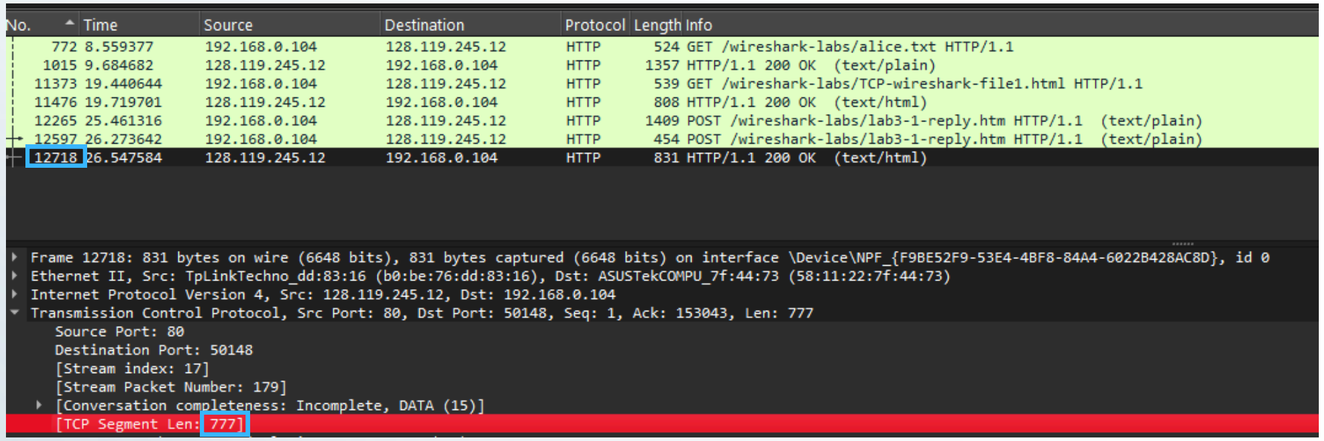
EstimatedRTT = (1- 0.125) × 0.273942 + (0.125) × 0.550817

EstimatedRTT = 0.239199 + 0.068852 = 0.308051

EstimatedRTT จะเป็น 0.3085521 วินาที

7. What is the length (header plus payload) of each of the first four data-carrying TCP segments?

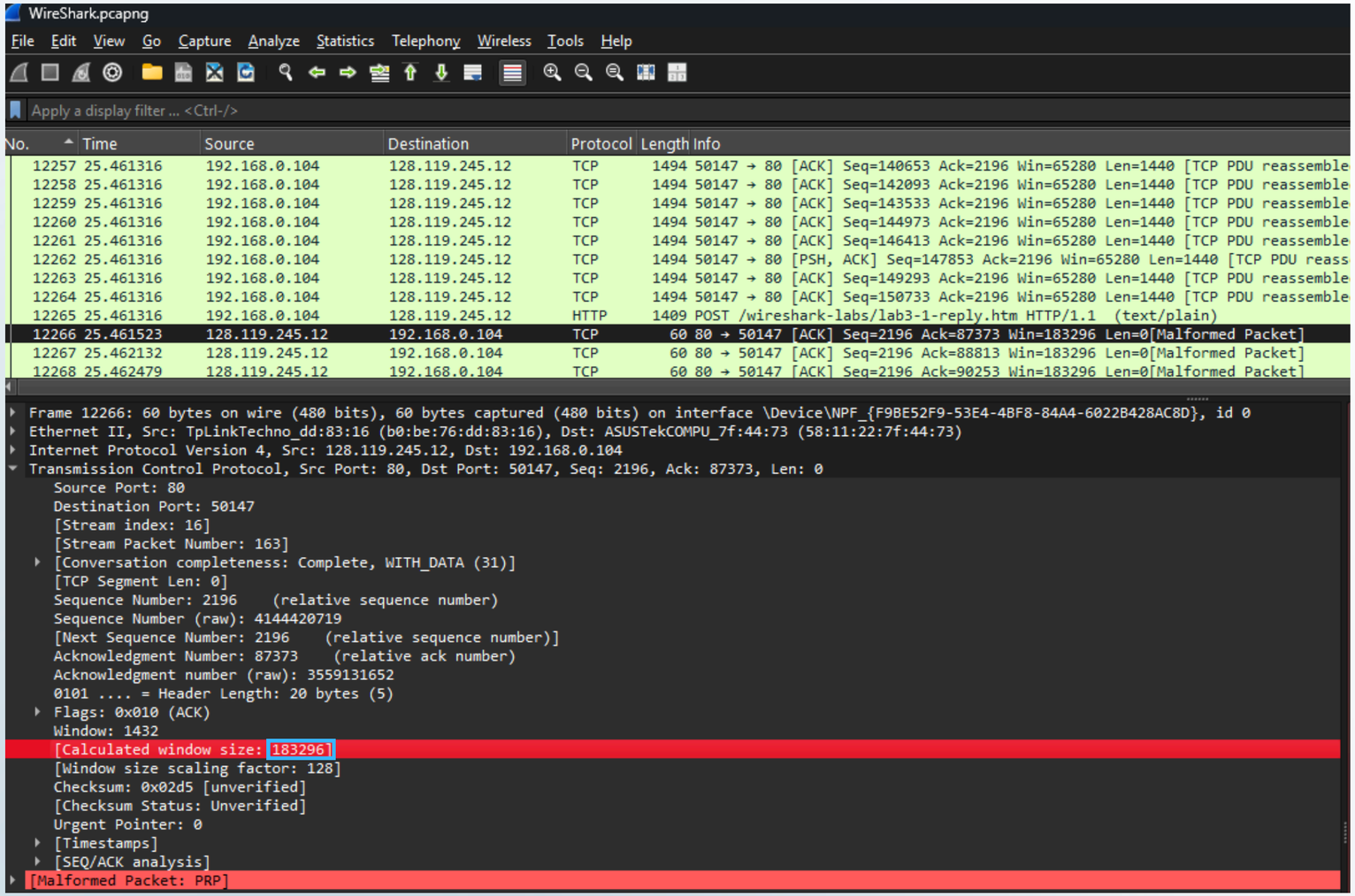




ความยาวของ TCP segment ตัวแรกคือ 400

ความยาวของแต่ละ TCP segment คือ 777

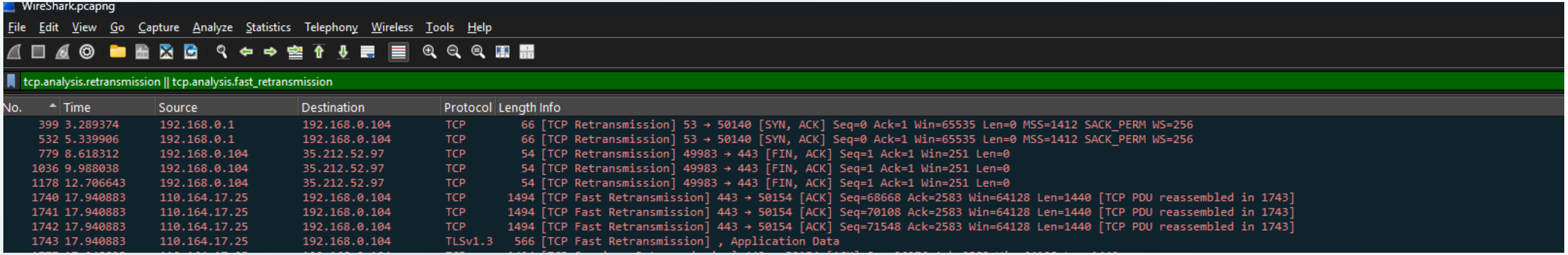
8. What is the minimum amount of available buffer space advertised to the client by gaia.cs.umass.edu among these first four data-carrying TCP segments ? Does the lack of receiver buffer space ever throttle the sender for these first four datacarrying segments?



Calculated window size: 183296

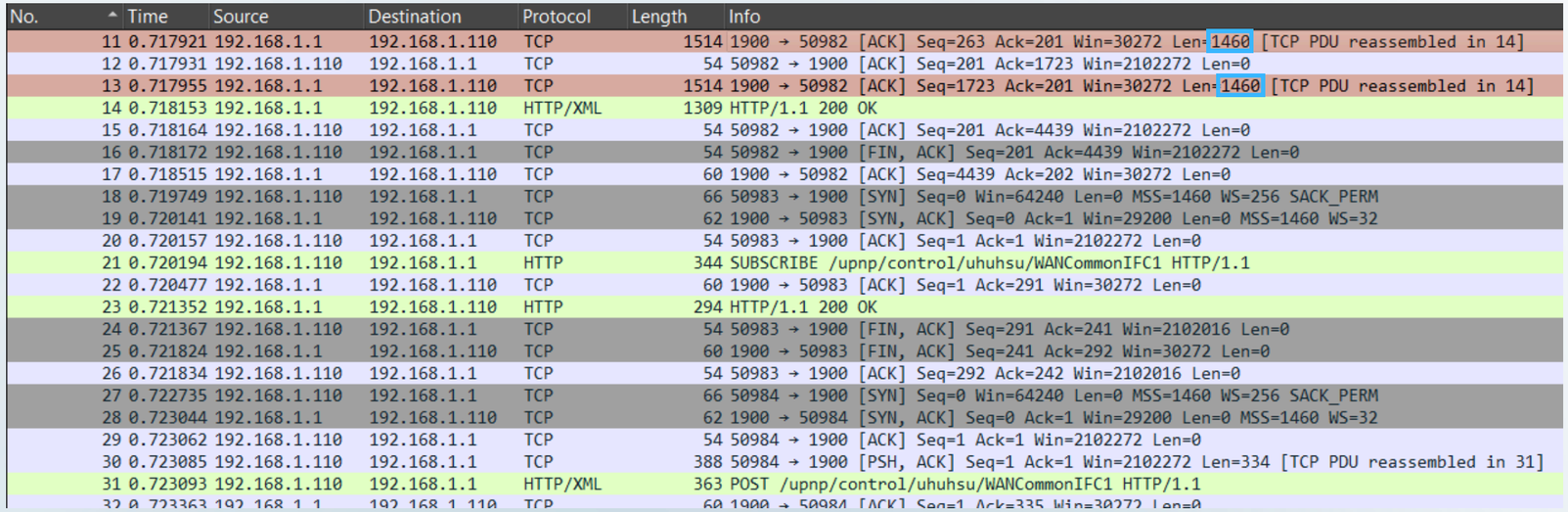
ผู้ส่งข้อมูลส่งได้ต่อเนื่องใน 4 เซกเมนต์แรกโดยไม่ถูกจำกัดความเร็ว

9. Are there any retransmitted segments in the trace file? What did you check for (in the trace) in order to answer this question?



มีแพ็กเก็ตที่ขึ้น [TCP Retransmission] หรือ [TCP Fast Retransmission] ในคอลัมน์ Info แสดงว่ามีแพ็กเก็ตที่ถูก retransmit

10. How much data does the receiver typically acknowledge in an ACK among the first ten data-carrying segments sent from the client to gaia.cs.umass.edu? Can you identify cases where the receiver is ACKing every other received segment (see Table 3.2 in the text) among these first ten data-carrying segments?



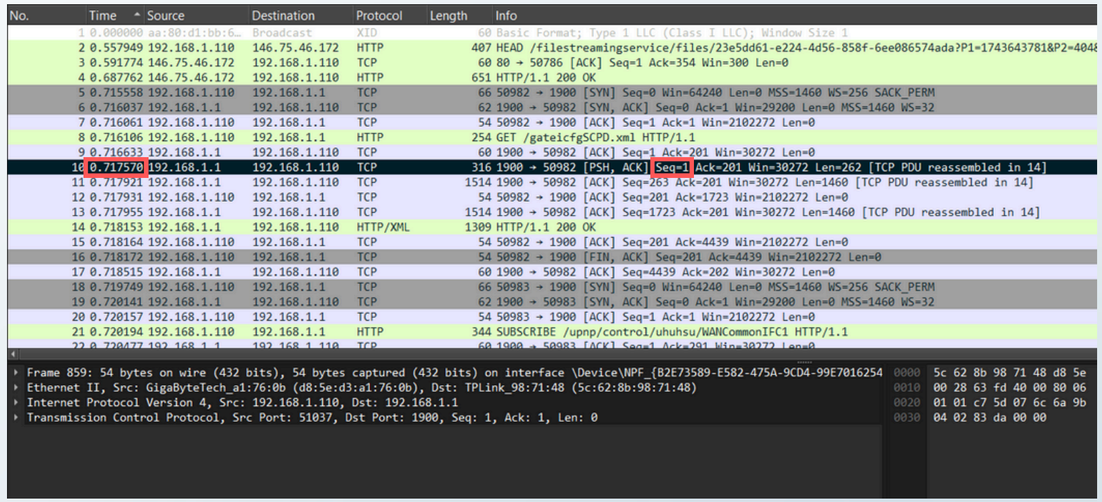
No.	Time	Source	Destination	Protocol	Length	Info
11	0.717921	192.168.1.1	192.168.1.110	TCP	1514	1900 → 50982 [ACK] Seq=263 Ack=201 Win=30272 Len=1460 [TCP PDU reassembled in 14]
12	0.717931	192.168.1.110	192.168.1.1	TCP	54	50982 → 1900 [ACK] Seq=201 Ack=1723 Win=2102272 Len=0
13	0.717955	192.168.1.1	192.168.1.110	TCP	1514	1900 → 50982 [ACK] Seq=1723 Ack=201 Win=30272 Len=1460 [TCP PDU reassembled in 14]
14	0.718153	192.168.1.1	192.168.1.110	HTTP/XML	1309	HTTP/1.1 200 OK
15	0.718164	192.168.1.110	192.168.1.1	TCP	54	50982 → 1900 [ACK] Seq=201 Ack=4439 Win=2102272 Len=0
16	0.718172	192.168.1.110	192.168.1.1	TCP	54	50982 → 1900 [FIN, ACK] Seq=201 Ack=4439 Win=2102272 Len=0
17	0.718515	192.168.1.1	192.168.1.110	TCP	60	1900 → 50982 [ACK] Seq=4439 Ack=202 Win=30272 Len=0
18	0.719749	192.168.1.110	192.168.1.1	TCP	66	50983 → 1900 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
19	0.720141	192.168.1.1	192.168.1.110	TCP	62	1900 → 50983 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1460 WS=32
20	0.720157	192.168.1.110	192.168.1.1	TCP	54	50983 → 1900 [ACK] Seq=1 Ack=1 Win=2102272 Len=0
21	0.720194	192.168.1.110	192.168.1.1	HTTP	344	SUBSCRIBE /upnp/control/uhuhsu/WANCommonIFC1 HTTP/1.1
22	0.720477	192.168.1.1	192.168.1.110	TCP	60	1900 → 50983 [ACK] Seq=1 Ack=291 Win=30272 Len=0
23	0.721352	192.168.1.1	192.168.1.110	HTTP	294	HTTP/1.1 200 OK
24	0.721367	192.168.1.110	192.168.1.1	TCP	54	50983 → 1900 [FIN, ACK] Seq=291 Ack=241 Win=2102016 Len=0
25	0.721824	192.168.1.1	192.168.1.110	TCP	60	1900 → 50983 [FIN, ACK] Seq=241 Ack=292 Win=30272 Len=0
26	0.721834	192.168.1.110	192.168.1.1	TCP	54	50983 → 1900 [ACK] Seq=292 Ack=242 Win=2102016 Len=0
27	0.722735	192.168.1.110	192.168.1.1	TCP	66	50984 → 1900 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
28	0.723044	192.168.1.1	192.168.1.110	TCP	62	1900 → 50984 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1460 WS=32
29	0.723062	192.168.1.110	192.168.1.1	TCP	54	50984 → 1900 [ACK] Seq=1 Ack=1 Win=2102272 Len=0
30	0.723085	192.168.1.110	192.168.1.1	TCP	388	50984 → 1900 [PSH, ACK] Seq=1 Ack=1 Win=2102272 Len=334 [TCP PDU reassembled in 31]
31	0.723093	192.168.1.110	192.168.1.1	HTTP/XML	363	POST /upnp/control/uhuhsu/WANCommonIFC1 HTTP/1.1
32	0.723363	192.168.1.1	192.168.1.110	TCP	60	1900 → 50984 [ACK] Seq=1 Ack=335 Win=30272 Len=0

ปกติ data ที่ acknowledges ได้จะอยู่ในช่วง 1448-1460 bytes ใน ack

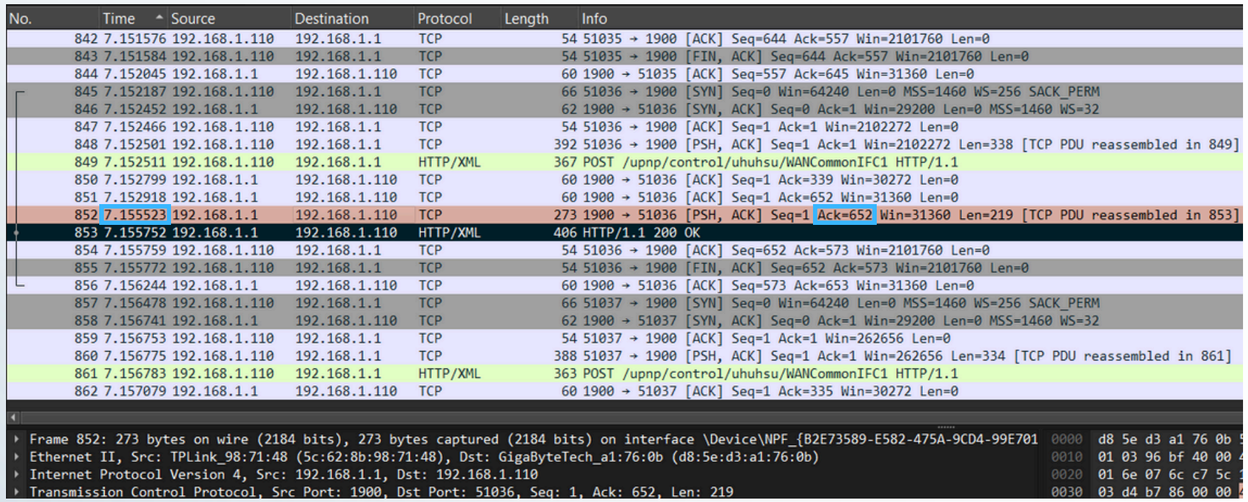
ครั้งนี้ได้ 1460 bytes ใน ack

11. What is the throughput (bytes transferred per unit time) for the TCP connection? Explain how you calculated this value.

TCP ตัวแรก



TCP สุดท้าย



TCP ของแรก: 1 bytes และตัวสุดท้ายที่ 652 bytes รวม 652 - 1 = 651 bytes

transmission time แรกที่ = 0.717570 วินาที

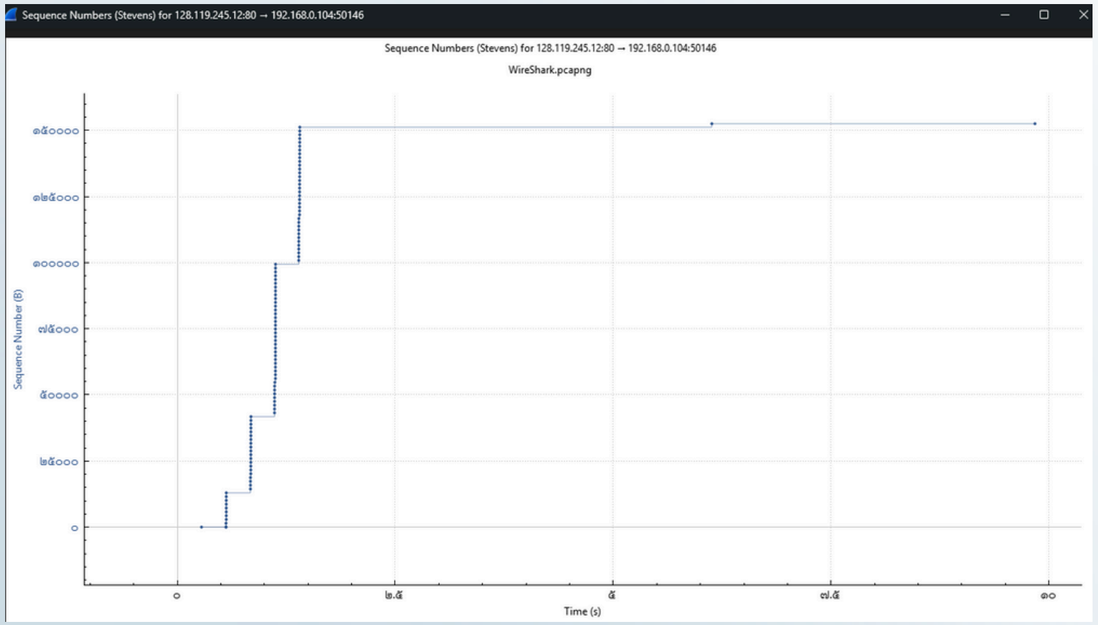
transmission time สุดท้ายที่ = 7.155523 วินาที

ต่างกัน 7.155523 - 0.717570 วินาที

ค่า throughput จะได้ 651/6.437953

= 101.1190979493016 MB/วินาที

12. Use the Time-Sequence-Graph(Stevens) plotting tool to view the sequence number versus time plot of segments being sent from the client to the gaia.cs.umass.edu server. Consider the “fleets” of packets sent around t = 0.025, t = 0.053, t = 0.082 and t = 0.1. Comment on whether this looks as if TCP is in its slow start phase, congestion avoidance phase or some other phase. Figure 6 shows a slightly different view of this data.



ช่วงแรก (t ≈ 0.025s) → Slow Start TCP ส่งแพ็คเก็ตเพิ่มขึ้นแบบทวีคูณ
ช่วงกลาง (t ≈ 0.053s - 0.082s) → อาจเป็นจุดเปลี่ยนไป Congestion Avoidance
การเพิ่มขึ้นของ Sequence Number เริ่มชะลอลง
ช่วงหลัง (t ≈ 0.1s) → Congestion Avoidance

13. These “fleets” of segments appear to have some periodicity. What can you say about the period?

จากพฤติกรรมของกราฟ คาบของกลุ่ม segments อยู่ที่ประมาณ 0.025s ถึง 0.05s ตามการเปลี่ยนแปลงระหว่าง Slow Start และ Congestion Avoidance

การคำนวณค่า TCP throughput ขึ้นอยู่กับการเลือกช่วงเวลาที่ใช้คำนวณค่าเฉลี่ย โดยทั่วไป เราจะใช้ระยะเวลาทั้งหมดของการเชื่อมต่อเป็นช่วงเวลาคำนวณ จากนั้นค่า throughput เฉลี่ยของการเชื่อมต่อนี้จะถูกคำนวณเป็นอัตราส่วนระหว่างปริมาณข้อมูลทั้งหมดและระยะเวลาการส่งข้อมูลทั้งหมด

14. Answer each of two questions above for the trace that you have gathered when you transferred a file from your computer to gaia.cs.umass.edu

ลำดับเฟสของ TCP: กราฟแสดง Slow Start (t ≈ 0.025s) ก่อนเปลี่ยนไป Congestion Avoidance (t ≈ 0.053s – 0.1s) ตามลำดับ

คาบเวลาของกลุ่มเซกเมนต์: อยู่ที่ประมาณ 0.025s – 0.05s, ตามการเปลี่ยนผ่านของ TCP

TCP Throughput: คำนวณจากอัตราส่วนของข้อมูลทั้งหมดต่อเวลาการส่ง